

A negative Ehrhart coefficient for the full face poset of a 13-gon

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

Let P_n be the full face poset of a convex n -gon, including its empty face and the polygon itself. We show that the Ehrhart polynomial $L_{13}(t)$ of its order polytope has linear coefficient $-298495711/35302608$. This gives a dimension-28 counterexample to the universal nonnegativity assertion in Question 6.4 of Lundström and Saud Maia Leite. The proof combines an exact transfer-matrix count, a counting proof of the polynomial degree, and a complete integer certificate. We make no minimality claim.

1 Statement and context

For an integer $n \geq 3$, write P_n for the full face poset of a convex n -gon, ordered by inclusion. It has $2n + 2$ elements. Define

$$\mathcal{O}(P_n) = \{x \in [0, 1]^{P_n} : x_u \leq x_v \text{ whenever } u \leq v\}, \quad L_n(m) = |m\mathcal{O}(P_n) \cap \mathbb{Z}^{P_n}| \quad (m \geq 0).$$

Here and below a dilation argument m is an integer. We also write $L_n(t)$ for the resulting Ehrhart polynomial and use $[t^j]$ for coefficient extraction in the ordinary monomial basis.

Theorem 1. *The order polytope of the full face poset of a convex 13-gon has dimension 28 and satisfies*

$$[t]L_{13}(t) = -\frac{298495711}{35302608} < 0. \tag{1}$$

Consequently, it is false that $[t^j]L_n(t) \geq 0$ for every $n \geq 3$ and every $0 \leq j \leq 2n + 2$.

Lundström and Saud Maia Leite ask precisely this nonnegativity question in [1, Question 6.4, p. 22]. Their notation $\mathcal{L}(P_n)$ denotes the face poset called P_n here. They prove nonnegativity of the order polynomial of the bare crown [1, Corollary 4.8] and report $[t]\Omega_{P_7}(t) = -3/1430$ for the full face poset [1, p. 22]. The latter is a different coefficient: $\Omega_P(r)$ counts maps into $\{1, \dots, r\}$, whereas $L_n(t) = \Omega_{P_n}(t+1)$. Theorem 1 concerns the Ehrhart polynomial itself. As adjacent context for exact certificates of failed polynomial properties associated with posets, Zeng [2] studies real-rootedness of a preorder support polynomial. The coefficient calculation below is self-contained.

2 Counting the lattice points

Dimension and polynomial degree

Denote the bottom and top of P_n by $\hat{0}$ and $\hat{1}$, its vertices by v_i , and its edges by e_i , with indices modulo n . Its covers are

$$\hat{0} < v_i, \quad v_i < e_i, \quad v_{i+1} < e_i, \quad e_i < \hat{1}.$$

Assigning the values $1/5, 2/5, 3/5, 4/5$ to the four successive ranks satisfies every nontrivial defining inequality strictly. Thus the polytope is full-dimensional. It is also integral: for any feasible x , each threshold vector $(\mathbf{1}_{x_p \geq a})_{p \in P_n}$, $0 < a \leq 1$, is a feasible zero-one vector. Partitioning $(0, 1]$ at the coordinate values expresses x as a convex combination of these vectors. Hence $\mathcal{O}(P_n)$ is the convex hull of its feasible zero-one vectors.

The degree bound needed for interpolation has an elementary counting proof. For a nonempty d -element poset P , let $a_k(P)$ count the weakly order-preserving surjections $P \rightarrow \{1, \dots, k\}$. A weakly order-preserving labeling into $\{0, \dots, m\}$ is specified by its k distinct values and the surjection obtained by ranking them. Therefore

$$L_P(m) = \sum_{k=1}^d a_k(P) \binom{m+1}{k}. \quad (2)$$

This is a polynomial of degree at most d . A linear extension gives $a_d(P) > 0$, so its degree is exactly d . In particular, $\deg L_{13} = 28$; the values at $0, \dots, 28$ determine it uniquely.

The transfer matrix and the two extremal faces

Remove $\hat{0}$ and $\hat{1}$ temporarily. Given vertex labels $x_0, \dots, x_{n-1} \in \{0, \dots, s\}$, the edge e_i has $s + 1 - \max(x_i, x_{i+1})$ possible labels. The edge choices are independent. Set

$$T_s[i, j] = s + 1 - \max(i, j) \quad (0 \leq i, j \leq s).$$

The number of labelings of this bare crown is consequently

$$C_n(s) = \sum_{x_0, \dots, x_{n-1}=0}^s \prod_{i=0}^{n-1} T_s[x_i, x_{i+1}] = \text{tr}(T_s^n), \quad x_n = x_0. \quad (3)$$

The vertices and edges are labeled; there is no quotient by rotations.

For the full poset, fix the bottom label a and top label b , where $0 \leq a \leq b \leq m$. Subtraction of a identifies the intermediate labels with a crown labeling in $\{0, \dots, b-a\}$. For each $s = b-a$ there are $m-s+1$ choices of a . Thus

$$L_n(m) = \sum_{s=0}^m (m-s+1) \text{tr}(T_s^n). \quad (4)$$

Every labeling occurs once. The cases $a = b$ are included, and $T_0 = (1)$ gives $L_n(0) = 1$.

3 The exact certificate

Let $\Delta f(m) = f(m+1) - f(m)$ and $q_k = \Delta^k L_{13}(0)$. Evaluation of (4) at $m = 0, \dots, 28$ gives the integers in Table 1. A complete program for this evaluation appears in Appendix A; the data supplement also records all 29 input counts. By (2) and Newton interpolation,

$$L_{13}(t) = \sum_{k=0}^{28} q_k \binom{t}{k}. \quad (5)$$

For $k \geq 1$, taking the linear term of $t(t-1)\cdots(t-k+1)/k!$ gives $[t]\binom{t}{k} = (-1)^{k-1}/k$. Put $D = \text{lcm}(1, \dots, 28) = 80313433200$. The entries of the table satisfy the integer identity

$$D[t]L_{13}(t) = \sum_{k=1}^{28} (-1)^{k-1} q_k \frac{D}{k} = -679077742525. \quad (6)$$

Since $\gcd(679077742525, 80313433200) = 2275$, reducing this fraction proves (1), and hence Theorem 1.

Table 1: The complete Newton certificate $q_k = \Delta^k L_{13}(0)$.

k	q_k	k	q_k
0	1	15	10987182053666969367338446
1	271444	16	22663756250271659749496179
2	1385407029	17	38626371277471350363061136
3	873183700042	18	54495716450410990966519984
4	152676439727865	19	63603815164255291057739776
5	11130331724443386	20	61202929859909033692840448
6	425668126056497142	21	48235969872663281855610880
7	9832718437409692188	22	30809277997317410488827904
8	150447992992033048364	23	15694300652256494183645184
9	1624674734818696934464	24	6224441141787228077752320
10	12956253841705283993234	25	1851954783159309833338880
11	78875167386815008012596	26	388821936255030835281920
12	375751539714927755376465	27	51359667879563260592128
13	1427065565567816253447394	28	3209979242472703787008
14	4381516440327096729272081		

Scope and reproducibility. The computation uses arbitrary-precision integers and exact rational arithmetic throughout. In addition to dense matrix powers, separate implementations using prefix/suffix sums and inverse tridiagonal determinants reproduced all 29 defining counts and all 29 Newton entries. These checks supplement the counting argument and the theoretical degree bound; a vanishing extra difference is not used as a degree proof. The theorem gives a counterexample at $n = 13$. It does not determine the smallest such n , the smallest dimension among all order polytopes, or the behavior for all larger n . The literature comparison is confined to the cited sources, and no priority claim relative to later work is made.

Computational and AI assistance. OpenAI Codex assisted with the counting argument, exact-arithmetic programs, certificate verification, and manuscript preparation. A separate model-context pass checked the proof and recomputed the certificate using tridiagonal determinants and exact Lagrange interpolation. This was an additional computational review, not external peer review or formal proof-assistant verification.

A A complete verification program

The following Python 3 program uses only the standard library and performs no file writes. Reversing both matrix index orders changes T_s into $A_s[i, j] = \min(i+1, j+1)$, which has the same power traces. Run `python3 verify.py` alongside this article's source files. It prints all counts, the table entries, and the integer and rational certificates. The recorded computations used Python 3.9.6.

```

from fractions import Fraction
from math import comb, gcd

def trace13(s):
    size = s + 1
    a = [[min(i + 1, j + 1) for j in range(size)]
          for i in range(size)]
    columns = list(zip(*a))
    power = [[int(i == j) for j in range(size)]
              for i in range(size)]
    for _ in range(13):
        power = [[sum(x*y for x, y in zip(row, col))
                  for col in columns] for row in power]
    return sum(power[i][i] for i in range(size))

def certificate():
    traces = [trace13(s) for s in range(29)]
    counts = [sum((m-s+1)*traces[s] for s in range(m+1))
              for m in range(29)]
    q = [sum((-1)**(k-j)*comb(k, j)*counts[j]
              for j in range(k+1)) for k in range(29)]
    denominator = 1
    for k in range(1, 29):
        denominator *= k // gcd(denominator, k)
    numerator = sum((-1)**(k-1)*q[k]*(denominator//k)
                    for k in range(1, 29))
    coefficient = Fraction(numerator, denominator)
    assert (denominator, numerator) == (80313433200, -679077742525)
    assert coefficient == Fraction(-298495711, 35302608)
    return counts, q, denominator, numerator, coefficient

if __name__ == "__main__":
    counts, q, denominator, numerator, coefficient = certificate()
    print("counts =", counts)
    print("q =", q)
    print("D =", denominator, "S =", numerator)
    print("[t]L_13(t) =", coefficient)

```

References

- [1] T. Lundström and L. Saud Maia Leite, *Order polytopes of crown posets*, European Journal of Combinatorics **133** (2026), 104304. doi:10.1016/j.ejc.2025.104304.
- [2] Z. Zeng, *An Eight-Element Counterexample to Real-Rootedness of Preorder Support Polynomials*, 2026. doi:10.2139/ssrn.7385158.